Solution Architecture

Date	03 July 2026
Team ID	xxxxxx
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer

Web Application (HTML, CSS, JavaScript)
Responsive User Interface
Crop Recommendation Dashboard
Prediction Result Page

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API Gateway / Load Balancer

Flask REST API
HTTP Request Handler
Input Validation
Prediction Request Routing

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Auth Service	**Core Logic Service**	**External API Integrations**
No Login Required Public Access Web Application Input Validation & Security Checks	Crop Recommendation Engine Random Forest ML Model Data Preprocessing Feature Scaling	Future Weather API Future Soil API Future Fertilizer

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Data / Storage Layer

Crop Recommendation Dataset
CSV Dataset
Trained Model (.pkl)
Prediction Logs

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	Provides a user-friendly interface where farmers enter soil and weather parameters to receive crop recommendations.	HTML, CSS, JavaScript
[API Gateway]	Receives user requests, validates inputs, and forwards them to the prediction engine.	Flask REST API
[Microservices]	Processes input features, loads the trained Random Forest model, performs crop prediction, and generates recommendations.	Python, Scikit-learn, Pandas, NumPy
[Database]	Stores the crop dataset, trained ML model, and prediction-related files.	CSV Dataset, Pickle (.pkl), Local Storage